



Frankfurt University of Applied Science High Integrity System
Safety Critical Computer System

Fall Detection System

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Overview of the System

This project aims to develop a fall detection system using smartphone sensors. The system collects real-time data from the accelerometer and gyroscope sensors, processes it to detect falls, and uses a graphical user interface (GUI) to alert users. Key features include accurate and timely detection, high reliability to prevent false alarms, and thorough testing for flexibility following the V-Model XT. Our fall detection system continuously reads data from the smartphone's sensors and triggers alerts if a fall is detected, ensuring help can be summoned promptly if the user does not respond.

Introduction

The objective of this project is to develop a fall detection system using smartphone sensors. The system continuously reads data from the smartphone's accelerometer (Ac) and gyroscope (Gy) sensors, capturing a data reading every five seconds. If a potential fall is detected based on this sensor data, the system triggers a wait period of ten seconds, during which the user can respond to the alert. If no response is received within this window, the system automatically sends an emergency notification to the user's designated contacts, ensuring prompt assistance.

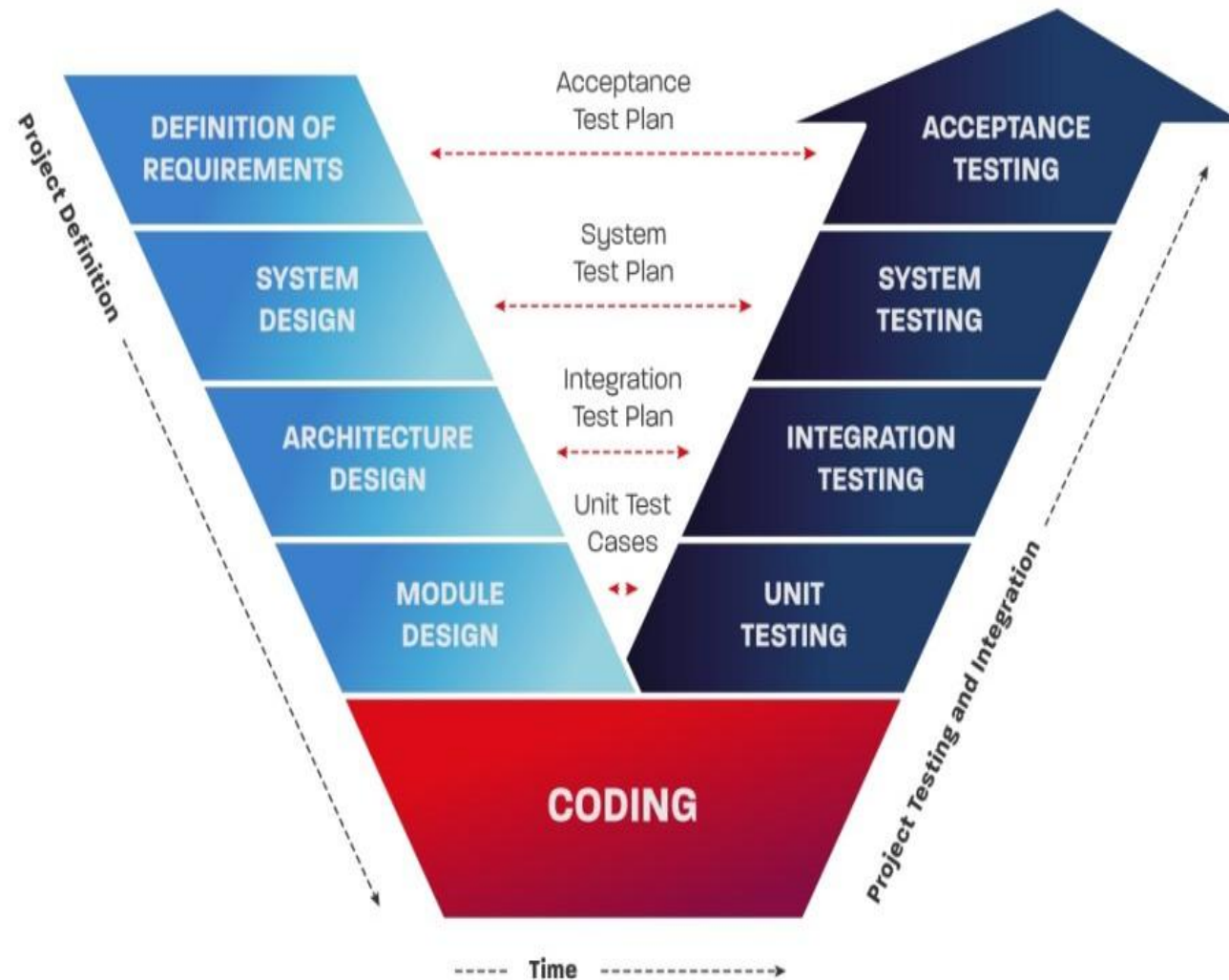
➤ Continuous Monitoring

This continuous monitoring system allows individuals to maintain their independence while providing a safety net that enhances their overall well-being.

➤ Leveraging Smartphone Sensors

By leveraging the sensors already present in smartphones, the system aims to reduce the incidence and severity of fall-related injuries.

Methodology



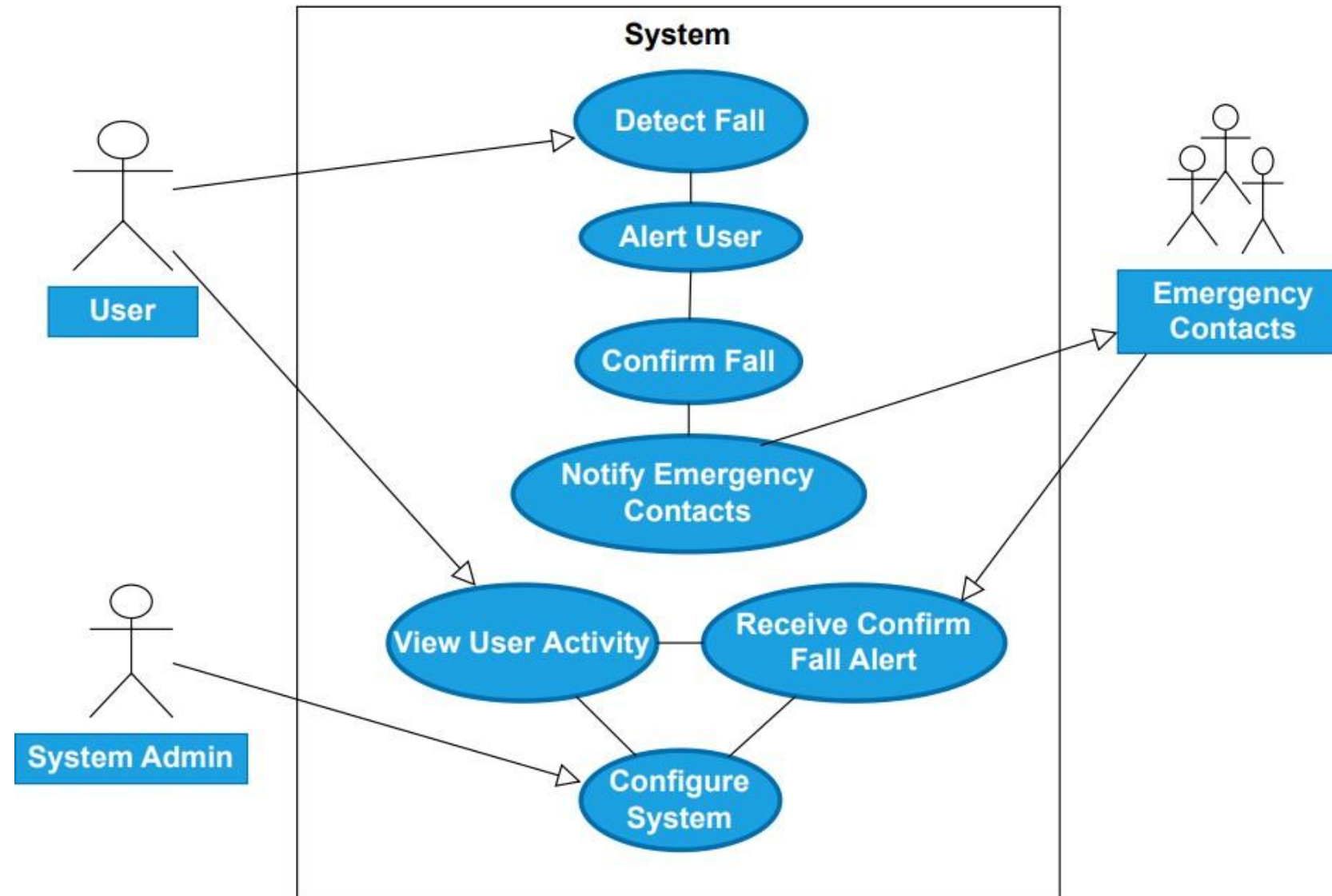
➤ Process Model: V-Model XT :

- Transparency: Document-centric approach for clear communication.
- Consistency: Thorough testing at each development phase.

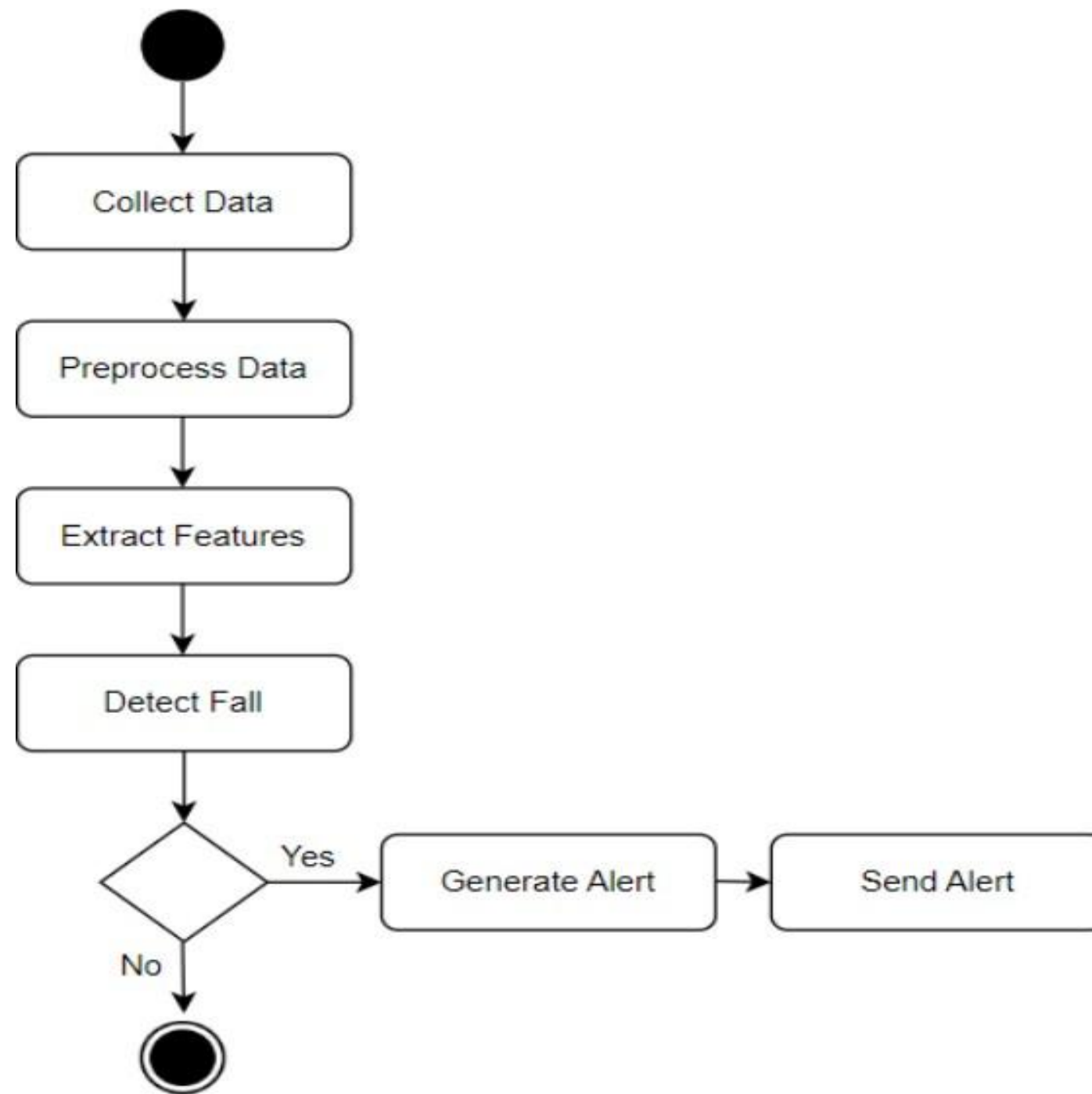
Phases:

- Requirement Analysis
- System Design
- Implementation
- Testing
- Maintenance

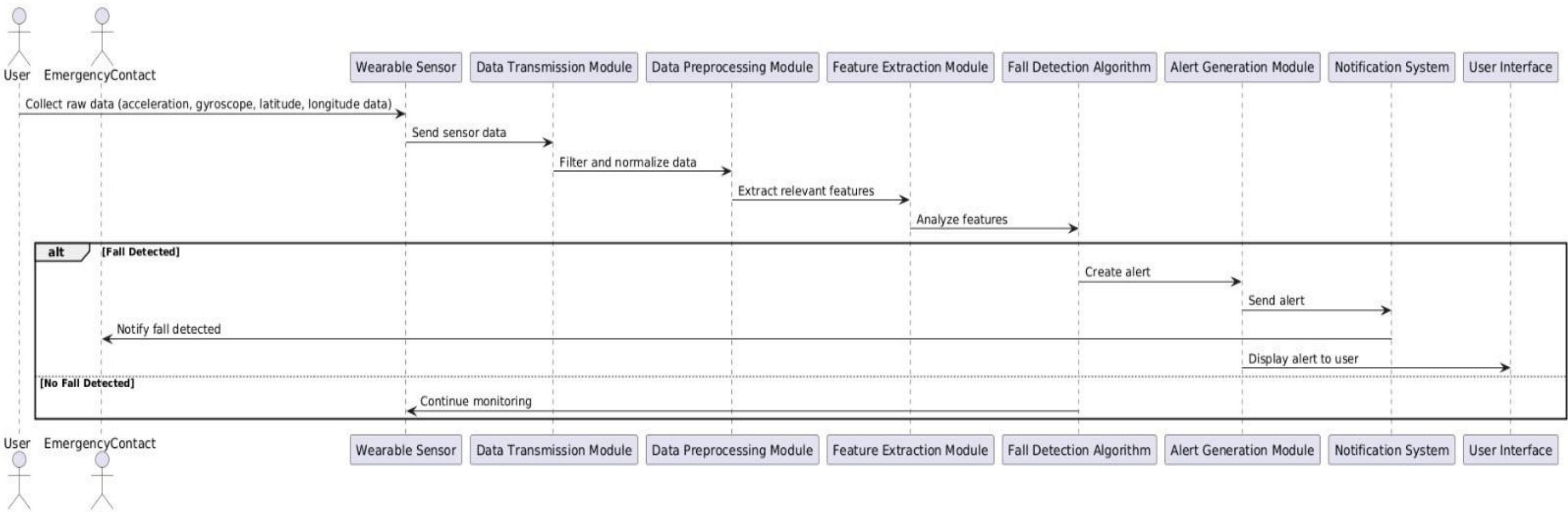
System Design



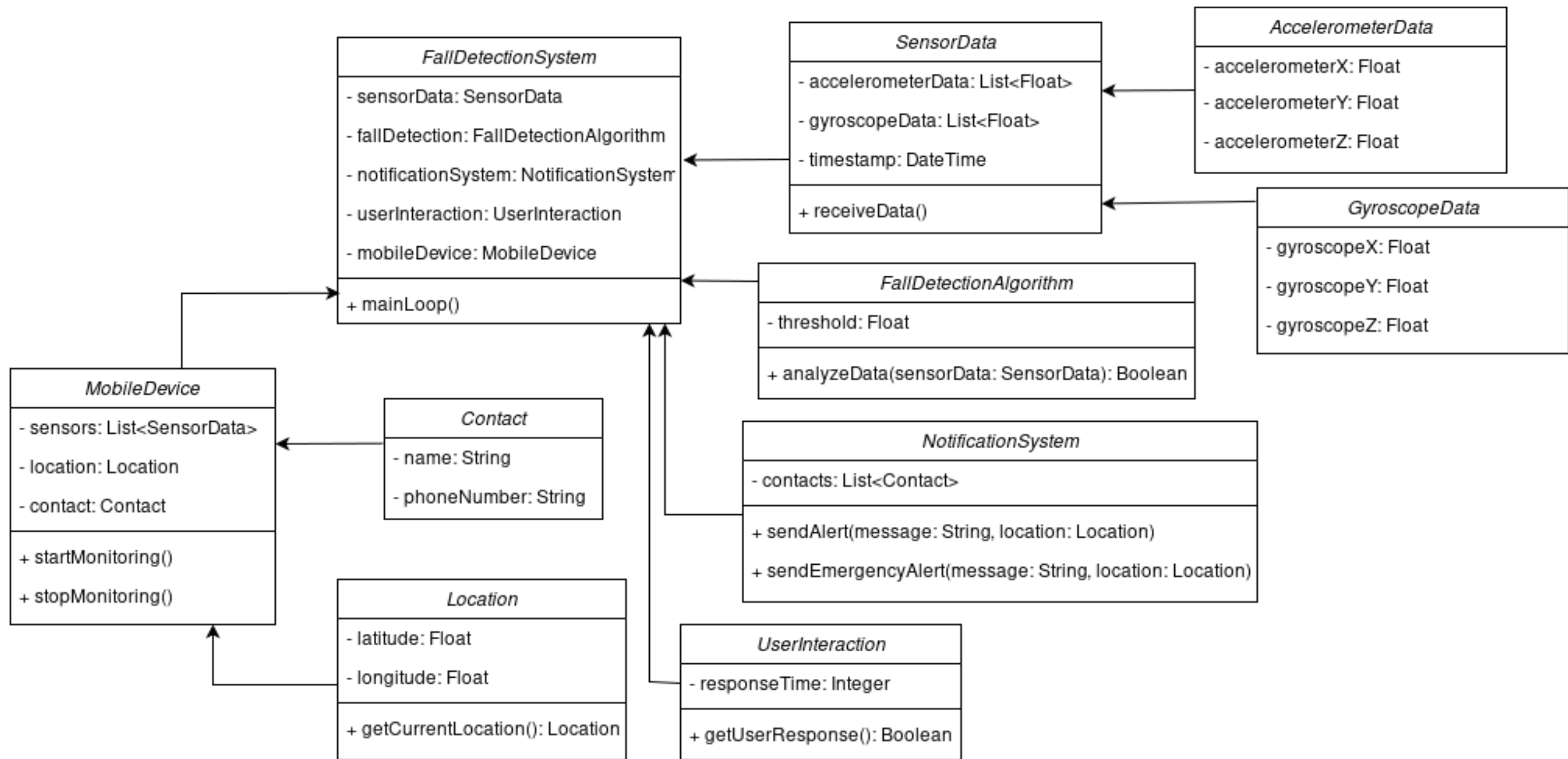
Use case Diagram



Activity Diagram

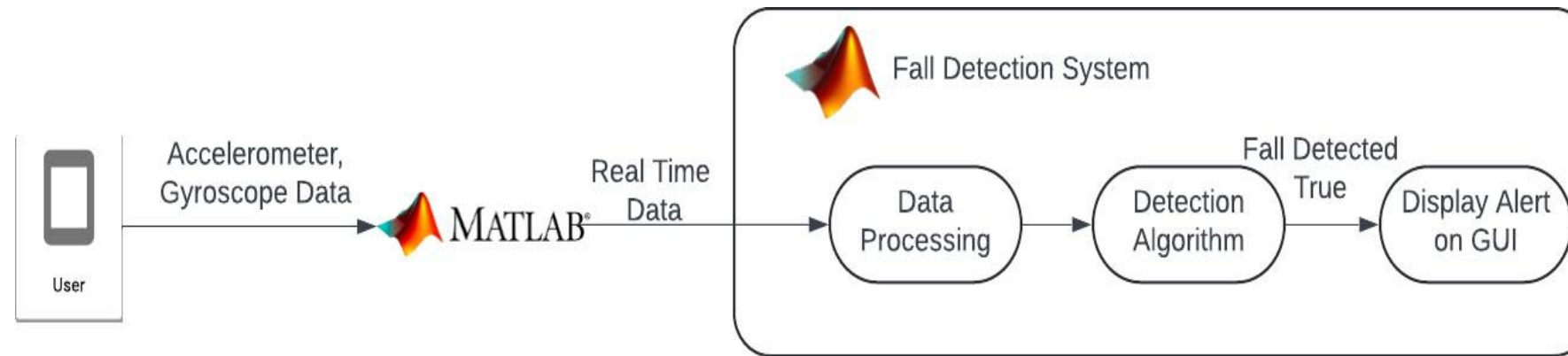


Sequence Diagram



Class Diagram

Architectural Design



The architecture of the Fall Detection System is structured in the following manner: The process starts with a user using a mobile device equipped to capture accelerometer and gyroscope data. This data is sent to MATLAB Mobile, which acts as the interface for real-time data input into the MATLAB-based detection system. Within MATLAB, the data undergoes processing and is then fed into a specialized algorithm designed for fall detection. Upon detecting a fall, the algorithm generates a true value, triggering an alert on a graphical user interface (GUI). The final component encompasses the entire MATLAB-based detection system, illustrating the sequence from data processing to alert activation.

Sensors

Crucial for fall detection by providing data on movement and orientation.

➤ Accelerometer:

- Measures linear acceleration along X, Y, and Z axes.
- Distinguishes between static forces (gravity) and dynamic forces (movement/vibration).

➤ Gyroscope:

- Measures rotational velocity around X, Y, and Z axes.
- Tracks changes in orientation and rotation.

Threshold Method and Mathematical Model

Threshold Method:

Differentiates normal activities from potential falls using predefined thresholds.

Mathematical Model:

- Data Collection
- Magnitude Calculation:

$$\text{Acceleration Magnitude: } |a| = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

$$\text{Angular Velocity Magnitude: } |\omega| = \sqrt{\omega_x^2 + \omega_y^2 + \omega_z^2}$$

- Threshold Values:

Acceleration Threshold: $T_a = 2g$

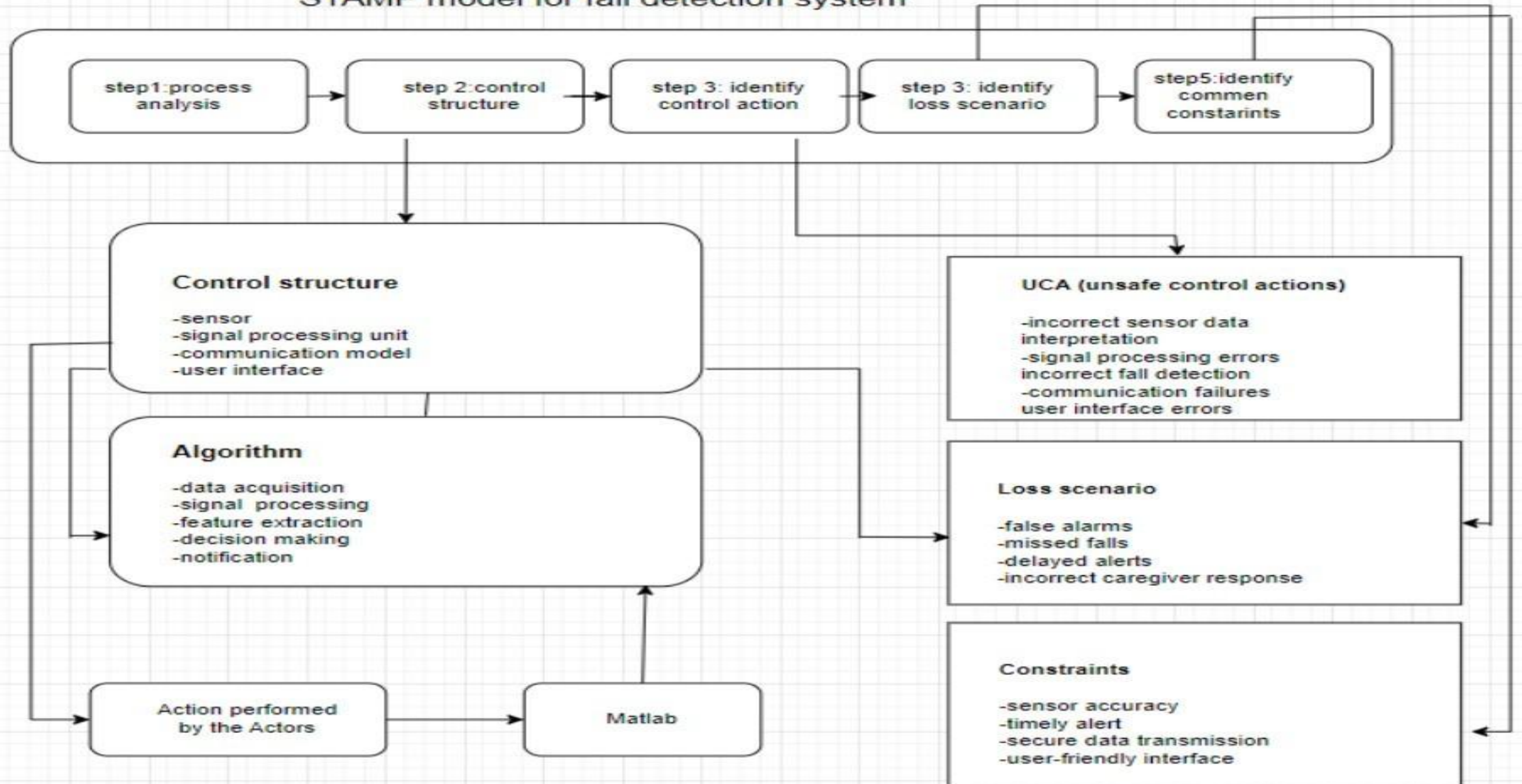
Angular Velocity Threshold: $T_\omega = 300 \text{ deg/}$

- Threshold Comparison:

A fall is detected if $|a| > T_a$ or $|\omega| > T_\omega$

STAMP ANALYSIS

STAMP model for fall detection system



Result

2 12:19 PM

Sent from your Twilio trial account -
Fall detected! Location: <https://www.google.com/maps?q=50.131346,8.692257>

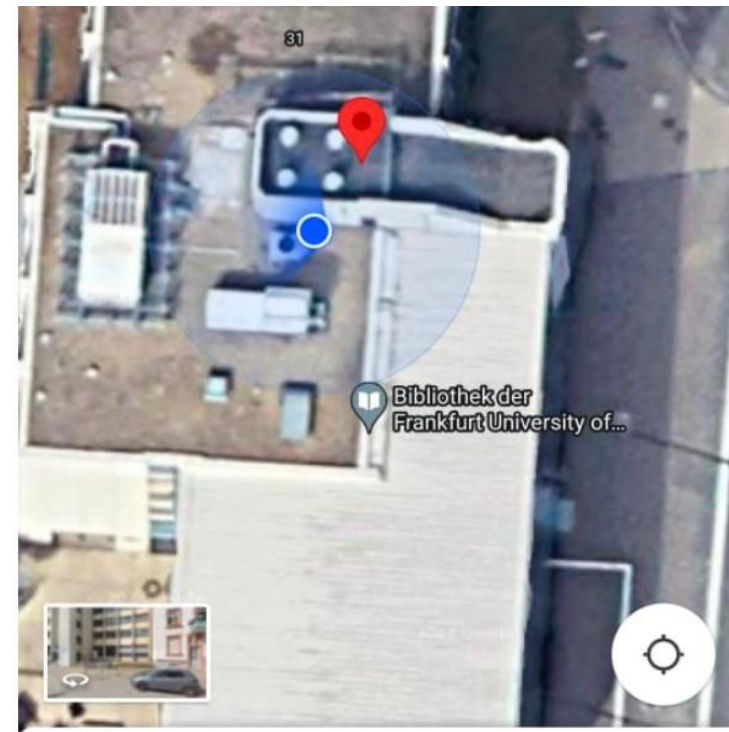
Sent from your Twilio trial account -
Fall detected! Location: <https://www.google.com/maps?q=50.131334,8.692279>

This message is sent from an unknown number.
 Identify Sender ID automatically



Text message

2



Kleiststraße 31



 Save

 Share

 Add label

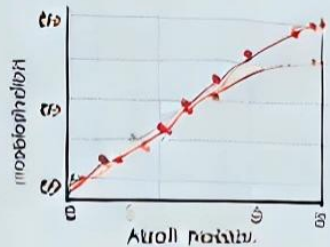
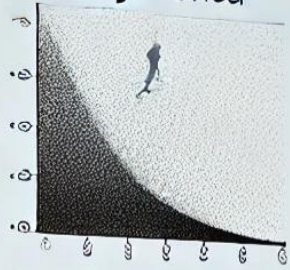
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Conclusion

- Disturbances from external sources
- from external sources
- from external sources
- from external sources
- from external sources



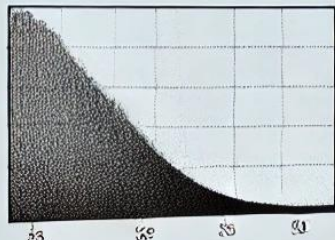
Key Method



Acceleration Magnitude



Threshold Method



- Acceleration magnitude
- AXA-2 -- AY22
- AX -- 23 30222

Angular Velocity magnitude Angular Velocity threshold



- 70 - 337/S
- 300 deg / S3

Angular Velocity



Conclusion

Summary:

- Objective Achieved: Developed a reliable fall detection system.
- Methodology: Utilized V-Model XT for structured development and testing.

Effectiveness:

- High Accuracy: Effective in detecting falls and sending timely alerts.
- Enhanced Safety: Prompt emergency notifications to designated contacts.

Challenges:

- False Positives/Negatives: Balancing accuracy.
- Usability: Ensuring ease of use for elderly users.
- Data Privacy: Safeguarding user data.

Future Steps:

- Wearable Integration
- Mobile Application Development
- Additional Features: Voice commands, enhanced notifications, battery optimization.